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American Connector Company (A)

"Looks bleak." Those were the words Denise Larsen, the Vice President of Operations at American Connector Corporation said calmly as she read the report on a competitor's plans to build a new electrical connector plant in the United States. The competitor, DJC Corporation of Japan, had become a dominant supplier of electrical connectors in its home market in recent years, after building what was rumored to be one of the most efficient connector plants in the world. However, despite its success in Japan, DJC was barely a contender in the U.S. market. The company had no plants in the U.S. and only a small sales force there. Larsen knew that this could all change quickly. As she explained to her assistant, Jack Mitchell, a recently graduated M.B.A.:

> There have been rumors the last few years that DJC would build a new plant here to launch an attack on the U.S. market. But with the market so crowded with competitors and burdened with excess capacity, no one took them seriously here. Either way, we figured that we still had a cost advantage. But if your report is right, and if DJC can operate a plant here like the one they have in Japan, I think DJC could quickly grab some market share here.

Larsen was worried because she felt American's position was particularly vulnerable at the moment. She was chiefly concerned with the company's connector plant in Sunnyvale, California since it had been struggling with a series of operating problems during the past year. Costs at Sunnyvale were increasing while quality seemed to be deteriorating. In the past month, she and Andrew Li, the new plant manager there, had discussed ways to improve the plant's performance. Now she wondered whether the DJC situation called for a completely new manufacturing strategy.

The Electrical Connector Industry in the Early 1990s

Electrical connectors were devices made to attach wires to other wires, attach wires to outlets, attach wires, components or chips to PC boards, or attach PC boards to other boards. A connector typically had two main body parts—a plastic housing and metal socket pins or terminals. The housing was usually made of a plastic resin such as a polyester, nylon, or polycarbonate. The metal pins could be made and plated with a number of different metals, ranging from tin to gold. **Exhibit 1** illustrates several basic product designs.

The connectors were used in a variety of product applications, including military and aerospace electronics, industrial electronics, telecommunications equipment, computer and office equipment, automobiles, and consumer electronics and appliances. Each application—often each

Professor Gary Pisano prepared this case with assistance from Research Associate Sharon Rossi. *The case was prepared as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation. Data have been disguised for purposes of confidentiality.*

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producer—called for different connector specifications. In 1990, there were over 700 standard connector product lines in North America alone. Standard designs were those which had been established by the International Institute of Connectors and Interconnect Technology (IICIT), the National Electronics Distributors Association (NEDA), or by end use industries. Other designs were custom-produced, usually on a one-year contract with a single connector company and were often accepted as industry standards after the contract expired. Since each type of connector had its own set of specifications, suppliers which produced many types of connectors (sometimes hundreds or thousands of models) for different industries were finding it difficult to meet the increasing number of specifications. In 1992, attempts were made to standardize product specifications among the many different industry associations.

Because connector types were made of different materials and varied from low- to high-technology, they varied dramatically in price, as well. For example, a simple connector such as a phone jack sold for only a few cents, whereas a custom designed connector such as one used in military electronics sold for several dollars. The cost of connectors used in any product typically counted for 2% or less of the cost of the end product.

In the 1970s, the U.S. connector industry[1] had experienced very rapid growth as firms built up capacity to meet the growing demand (particularly for computer applications). But when demand slowed in the mid to late '80s, there were too many suppliers and too much capacity. Price competition intensified as more offshore producers entered the U.S. market.

In the 1990s, the U.S. connector industry was characterized as a hostile environment. There were more than 900 suppliers and sales continued to slacken. In 1991, sales were down 3.9% from the previous year, while the ten industry leaders were on average down 7.9%. The abundance of suppliers gave customers leverage to demand reduced prices, improved quality, and faster delivery. At the same time, many customers were working to reduce the number of suppliers they did business with. OEMs that previously dealt with 10 or 12 connector vendors had reduced their suppliers to as few as four. Pressures spurred a trend of mergers and acquisitions in the industry and analysts predicted that the number of connector suppliers in the U.S. might drop to 400 or fewer by the end of the 1990s.

Electrical connectors were very engineering intensive products and were critical to product performance. As electronic circuitry became more miniaturized and operated at higher speeds, new connectors had to meet more demanding requirements for space, weight, cost, quality, reliability and performance.

In 1991, worldwide sales of interconnect products totaled roughly $16 billion. The top ten worldwide leaders accounted for $6.67 billion, but the total industry (1,200 competitors) was very fragmented. The dominant company in the industry was AMP, Inc. It held 16% market share with sales of $2.6 billion in 1991. The second tier of companies consisted of six other companies, each of which had sales in the $500 million to $800 million range. DJC and American Connector were among the second tier companies worldwide. Companies in the third tier had sales in the $250 to $500 million range. In total, there were 28 firms with sales greater than $100 million.

[1]The "connector industry" included three types of interconnect products: connectors, cable assemblies and backpanels. While American Connector and DJC Corporation produced all types of interconnect products, the Sunnyvale and Kawasaki plants manufactured only connectors.

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Profile of DJC Corporation

The DJC Corporation produced a variety of electrical connectors used in computers, telecommunications, and consumer electronics. For these applications, DJC produced four basic types of connectors: wire-to-wire, wire-to-outlet, item-to-board, and board-to-board.

There were several aspects characterizing DJC's competitive strategy in most of its product lines. First, the company cultivated and maintained close links with the major computer, telecommunications, and electronics companies and distributors in Japan. These relationships represented an important entry barrier in the Japanese connector market. Secondly, the company's design strategy emphasized simplicity and manufacturability over innovation. Early DJC product designs were based on reverse engineering of other companies' designs, including those of American Connector. As one former manager of DJC explained:

> In 1965, we copied other companies' products. Americans made good products and the U.S. market was the most advanced in the world. Our R&D was geared entirely toward analyzing U.S.-made products, copying improvements as they were made. By 1975, our quality was as good as theirs. After that, the production process became the basis for competition.

However, DJC's design strategy went beyond simply copying U.S.-made connectors. The company paid very careful attention to customer and user needs in adapting American designs to the particulars of the Japanese market. For example, DJC connectors were designed for maximum compactness since this feature was very important to Japanese OEMs, particularly those producing consumer electronics. DJC also adapted the designs to economize on raw materials (which were nearly twice as expensive in Japan as they were in the United States) and to simplify manufacturing. Features which did not add perceived value to customers (such as color-coded housings) were eliminated.

Finally, and perhaps most importantly, DJC viewed highly efficient manufacturing as absolutely critical to its competitive strategy. The company historically relied upon manufacturing as the major means to achieve their overall profit goals. As one former Managing Director of DJC described it:

> In electrical products, high quality is a prerequisite for success. With large established competitors fighting for a maturing market, a low cost position becomes necessary for long term success. Manufacturing excellence is the source of both and has therefore been at the heart of DJC.

The importance of manufacturing to DJC was reflected in the organization of the company (see **Exhibit 2**). For example, Mr. Okada, the head of production, was responsible for the operations of four domestic factories and reported directly to the company president, Mr. Esaka. In addition, the balance of power between manufacturing and the sales/marketing division was clearly tipped in favor of manufacturing. For example, sales/marketing had little power to alter production schedules, product mix or lead times. As one former manager at DJC explained: "Sales sometimes needs an unscheduled delivery, but manufacturing just does not allow it. There isn't even any debate."

DJC's President, Mr. Esaka, was considered to be a dominant, hands-on leader. He was hand-picked by DJC's founder to become the company's president in 1971. Many within DJC viewed Esaka as *the* decision maker within the company. As one former executive put it: "Our strategy is pure and strict, driven by Esaka himself. People may bicker back and forth...but everyone knows they are personally responsible to achieve the goals set out by the president."

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The Kawasaki Plant

During the early 1980s, the Japanese connector market experienced increased labor and raw material costs, a rising yen, and increased import penetration. Amidst these conditions, top executives at DJC were concerned that the company might not be able to maintain its historical 50% gross margins in the connector business. This concern led Mr. Esaka to formulate his vision of a plant which could achieve "the ultimate rationalization of mass production." His vision called for a highly automated, continuously operating plant which could meet the following three goals: First, the plant must achieve asset utilization of 100%. Secondly, yield on raw material must reach 99%. Finally, customer complaints could not exceed 1 per million units of output. While cost was not stated as an explicit goal, everyone understood that if the above three goals were met, the plant would be one of the lowest cost producers in Japan.

DJC chose to build its "ultimate" new plant in Kawasaki, Japan; the plant was completed and began operating in 1986. Management chose the Kawasaki site for several reasons. From a logistical point of view, Kawasaki offered the advantage of being located close to the major Japanese electronics companies. In addition, perhaps more importantly, Kawasaki was near the major raw material suppliers. This was particularly important because it was anticipated that most raw materials would be delivered from vendors on a daily or weekly basis. The Kawasaki area also had an ample supply of relatively young, highly skilled workers.

The Kawasaki plant was designed to produce a maximum of 800 million connectors per year, assuming 100% utilization. Initially, the plant produced only 80% to 90% of this volume. About 75% of this output was sold in Japan and 25% was sold in developing Asian markets outside of Japan. The plant operated 24 hours a day, seven days per week, 330 days per year. The main advantage of running the plant on a nearly continuous basis was that it avoided start-up and shut-down costs.

To successfully operate the plant on a continuous basis, as well as meet Esaka's yield and quality goals, Kawasaki management carefully integrated decisions and policies related to the product and process technology, workforce, production control, quality, and organization.

Plant Layout The Kawasaki plant was organized into four large cells, each of which was responsible for producing one of the four general types of connectors (wire-to-wire, wire-to-outlet, item-to-board, and board-to-board). With the exception of plating, all of the processes needed to manufacture a complete connector were located in each cell. Plating was organized separately in order to fully utilize the high fixed cost equipment and to protect the rest of the factory from exposure to corrosive chemicals and noxious fumes.

Each cell contained anywhere from two to six production lines, with each line consisting of terminal stamping, housing molding, assembly, and packaging. Each production line was responsible for producing a specific family of the cell's products. Successive processing stages in each line were located close to one another and in a straight line in order to minimize materials handling steps and to reduce as much as possible the distance work-in-process had to travel. For example, each plastic molding press in a cell was located only a few feet from the line's assembly operations. Operations were synchronized so that completed housing parts flowed almost continuously (via small bins) between each molding and assembly line. Because molding and terminal stamping equipment had shorter cycle times than assembly equipment, these processes were run below their top speeds in order to synchronize parts fabrication and final assembly. Assembly operations were almost completely automated. After assembly, connectors were inspected and transported a few yards to the cell's packaging area. In packaging, connectors were sealed individually in plastic on strips containing 2,000 units.[2] Each strip or "tape" was loaded onto a large reel. The central shipping

[2]Reels of 1,500 units were considered standard in the industry.

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department was responsible for packing each customer's order for shipment. The Kawasaki plant delivered to many customers on a daily basis and some of the largest customers received shipments every few hours.

Product Technology Product design reflected the goals of continuous and reliable operations and the need to economize on raw materials. Before commencing production at Kawasaki, product designs were thoroughly analyzed to determine ways in which the product might be made easier to manufacture and use less material. For example, product design of most connectors was standardized to reduce the number of product variations. In 1991, the plant produced only 640 different stock-keeping-units (SKU's), a relatively small number for a plant its size. The limited number of product variations, it was believed, reduced the costs and complexity associated with shorter production runs. To economize on the use of raw material, designers adapted some types of connectors to use pins plated with tin rather than gold. Though gold was the most reliable and durable material, tin was far cheaper and worked well in low power applications. To further simplify production and reduce costs, DJC packaged its connectors only on tape and reels. This packaging was particularly suited to customers with automated production environments, and it did not inconvenience customers with manual operations. DJC's engineers undertook extensive value-engineering to identify and implement cost saving design changes which did not compromise product quality or performance. These design changes are discussed in **Exhibit 3**.

Process Technology Process design activities reflected several basic principles. First, while the plant was to be highly automated, significant resources were devoted to what DJC called "pre-automation." Pre-automation referred to the activities required to make the production process suitable for highly reliable automation. It reflected the philosophy that a production process could only be automated after it was completely understood, properly designed, and properly laid out. To automate before might mean automating a process which was inherently inefficient or unreliable. During pre-automation, process flows were carefully analyzed to determine ways in which the process could be streamlined and inventories eliminated. Worker movements and motions were also studied to identify ways in which the process could be made more efficient. Pre-automation activities also included specifying raw material quality and process tolerance levels.

There were several examples of how pre-automation problem-solving affected the process. The warehouse was centrally located to simplify material flows and to economize on space. The amount of warehouse and floor space was intentionally limited so that there would be no room for excessive raw material or in-process inventories. To simplify material flows, each injection mold for plastic parts had a dedicated press and each press was dedicated to a single assembly line. Each assembly line was laid out in a continuous straight line from stamping to packaging. This made it possible for one operator to run two assembly lines. Only after the process had been "pre-automated" would steps be taken to implement automation.

The second principle guiding process design was the notion that it was better to use an old, reliable process than a new, less reliable one. Rather than taking chances with new technology, the plant relied on continuous improvement of existing proven processes. Reliable process technology was considered absolutely essential to keeping the process running smoothly, without inventory, on a continuous basis. To further ensure smooth runs, processes were generally operated below maximum speed.[3] Emphasis was also placed on maintaining equipment with the goal of eliminating unscheduled downtime.

A third principle guiding process design at Kawasaki was the emphasis on absolute reliability in upstream molding processes. According to one former DJC executive, "DJC views molding as the most critical part of the manufacturing process and has focused its efforts there." Its

[3]For example, an assembly line in the Kawasaki plant ran at 200 units/minute. By comparison, a similar line in American Connector's Sunnyvale facility ran at 500 units/minute.

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molding group included experts in polymer physics and many former employees of mold manufacturers. Several policies and decisions reflected DJC's desire to make molding virtually faultless. Molds were subjected to rigorous repair and maintenance schedules. For example, every mold received full maintenance every six months in addition to daily basic maintenance. Molds were also replaced relatively frequently to reduce the chance of failure and to allow mold technology to be upgraded. The average life of a mold at Kawasaki was three years. Taking into account purchase costs, maintenance costs, and costs of repair, Kawasaki's average annual cost per mold was $29,000.[4] The plant achieved mold yields in excess of 99.99%.

A fourth element of Kawasaki's approach to process technology was its reliance on in-house technology development. DJC's strong in-house process engineering competence was a result of both historical conditions and of strategic choice. One former employee described the historical conditions shaping the company's in-house process engineering expertise: "Because we were a small company and couldn't afford to buy equipment, we built a lot of it ourselves. There were extensive workshops in every factory."

During its early years, the company developed ideas about manufacturing by looking at the operations of emerging Japanese role models like Toyota. The decision to develop technology in-house was also strategically motivated. In Japan, contractual agreements with equipment vendors preventing resale to competitors were not common. Thus, DJC worried that its ability to achieve a competitive edge in process technology would be severely limited if it relied too much on equipment vendors. Indeed, part of this strategy was shaped by observing American Connector's experience with equipment vendors. As one DJC engineer recalled: "American Connector didn't develop their (connector) assembly machine themselves, they asked an equipment manufacturer to help them design the machine. That same equipment company then offered to sell us the identical machine." Thus, while the Kawasaki plant might buy standard equipment from vendors, it would make all proprietary design modifications in-house. In addition, Kawasaki designed all of its molds in-house and manufactured about half of them in-house as well (the most complex molds were all produced in-house). Kawasaki's goal was to eventually build 100% of its molds in-house.

A final element of Kawasaki's technology strategy was the inter-functional coordination of all its technology development activities. The plant's "Technology Development Division" was responsible for coordinating and managing the activities of the product planning section, the materials section, process engineering, and the molding technology group. Each section was assigned a specific set of goals. For example, it was the job of process engineering to design and modify new equipment and to identify opportunities for automation. However, it was the job of the Technology Development Division to ensure that all of these sections were working in concert to achieve a consistent set of explicit goals. These goals included: efficient resource utilization, design quality and manufacturability, smooth manufacturing introduction, shorter development cycle, and continuous process improvement.

One example of how this worked in practice was the development of a new resin to improve connector durability. The materials planning group solicited input from the product planning group on customer needs and requirements. Through this contact, the materials group learned that a more durable connector would help differentiate DJC's product in the customer's eyes. This led to a discussion with members of the R&D group about possible material breakthroughs which might create a more heat-tolerant and damage-resistant connector. At the same time, the materials group also held discussions with the process engineering group to learn about the potential new materials and the requirements of the manufacturing process. A new resin which improved connector durability emerged from this joint effort.

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[4]In comparison, American Connector's Sunnyvale plant spent approximately $40,000 per mold per year and molds averaged eight years of useful life.

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In the same way that the Technology Development Division coordinated various functions to improve product characteristics, it integrated the efforts of product planning, materials, process engineering and molding to reduce material costs and improve processes. It was estimated that if Kawasaki were to use American Connector's product design and packaging, material costs would total $20.90 per thousand units. However, because the Technology Development Division eliminated several design features of the product which used material but did not add value to the customer and implemented several waste reducing changes in the production process, the current Kawasaki material costs were only $14.89 per thousand units. This cost was still above American Connector Sunnyvale's $11.49 per thousand, but relevant raw materials were approximately twice as costly in Japan than in the United States. **Exhibit 3** contains an analysis of these improvements.

Sourcing Kawasaki maintained close relationships with a few suppliers of its key raw materials. These suppliers had to meet rigorous quality standards and were required to certify the quality of their products in every delivery. When Kawasaki received a shipment from a supplier, materials were used directly in production without further inspection. Suppliers were expected to further improve their quality over the long term. A manager of one of Kawasaki's Japanese suppliers commented: "Japanese customers will complain if the material causes problems in the manufacturing process and they meet with us frequently to discuss problems. DJC is our most demanding customer....They complain constantly." Quality improvement was considered a joint effort between Kawasaki and its suppliers. Any supplier which was having difficulty meeting specific requirements received technical support from Kawasaki.

Kawasaki's sourcing policy also demanded frequent delivery. Most of the raw materials (including resins for the housing, metals for the pins, and packaging) were delivered on a daily basis. The rest were either delivered weekly or monthly. Frequent deliveries allowed Kawasaki to maintain raw material inventories that averaged only five days.[5] The low level of raw material inventory, in turn, allowed Kawasaki have a relatively small warehouse and to reduce the amount of resources devoted to managing and controlling raw material stocks. While the procurement department relied on just-in-time deliveries, it used an M.R.P. program to plan longer term material requirements.

Quality Control The Q.C. Division pursued five objectives: improving quality control standards, improving the process inspection system, improving the precision of molded components, improving the quality of product designs, and reducing the plant's waste. Most of its work, which included modifying equipment, was done in conjunction with the Technology Development Division.

Production and Inventory Control To minimize yield and capacity losses associated with changeovers, production runs were scheduled to be as long as possible. On average, a production run of a particular model lasted one week, though some product lines were run on an almost continuous basis. Long runs were possible because of the limited number of SKU's produced at the plant. To further facilitate long runs, Kawasaki's plant had complete control over its schedule and mix and refused to make changes for unplanned orders.

As mentioned earlier, all aspects of the process and the plant layout were designed to achieve a smooth flow of materials and to minimize work-in-process inventory. With this design, management expected they would need to devote relatively few resources to inventory control. For the most part, this proved true: Kawasaki's processing lead times and work-in-process inventories each averaged two days.[6] However, Kawasaki maintained a relatively high finished goods inventory of 56 days.

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[5]In comparison, raw material inventories at American Connector's Sunnyvale plant averaged 10.8 days.

[6]Processing lead time was the time it took for a batch of connectors to be completed (i.e. placed in finished goods inventory) once the first processing operations on the batch were commenced in either molding or terminal stamping. Processing lead time included waiting time between operations.

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Workforce One of the explicit goals of the Kawasaki plant was to gradually reduce the number of direct production workers as well as support and overhead staff. Management expected that as the plant became increasingly automated, fewer direct production workers would be needed. Similarly, the number of employees dedicated to controlling various activities in the plant were also expected to decline as processes became more reliable, as inventories decreased, and as the overall complexity of the plant environment declined.

The general approach to selecting workers was to hire new graduates from high schools (for lower level positions) and from universities (for higher level positions). All employees hired into the Technology Development Division had to have a university degree. Kawasaki offered wages which were somewhat above the average for recent graduates in Japan. The goal was to attract people with generally high skills and aptitudes ("raw talent") who could be developed on the job. New hires received extensive training both through formal programs and through a job-rotation system designed to broaden their skill base. The average worker in Kawasaki shifted positions once every three years. Production workers were directly responsible for *all* activities affecting conversion and material flows and were not specialized to particular processes or functions.

While Kawasaki's wages were above average for new graduates, they were below average for experienced workers. Thus, while the compensation system and the opportunities for training tended to attract relatively qualified young workers, it tended to discourage workers from staying at Kawasaki for a long period of time. Many Kawasaki workers tended to leave the plant after the age of 35. The average employee stayed at Kawasaki for 9 years, while the average for large Japanese companies was 14.5 years.

Organization While Esaka was the driving force behind the Kawasaki plant, the plant operated under a high level of autonomy from corporate. Esaka set the goals, but left the plant manager completely free to pursue those goals in any way he chose. **Exhibit 4** provides an organizational chart of the Kawasaki plant.

The philosophy of pushing decision-making autonomy down in the organization followed at the plant as well. While the plant manager and his top staff were responsible for longer term planning issues facing Kawasaki, most tactical problems were solved by production employees. The number of employees dedicated to various control and support activities (such as accounting, production and inventory control, materials handling, etc.) was relatively low because all of the plant's technology and operating policies were designed to reduce the sources of problems that create a need for control staff. For example, a relatively continuous process flow eliminated much of the need for materials handling personnel. High process reliability made it unnecessary to employ mechanics for repair. Similarly, lack of work-in-process inventory greatly reduced the need for inventory control workers. Kawasaki's success in achieving this aim is reflected in data on the relative number of employees in various job functions (see **Exhibit 5** for a comparison of Kawasaki and Sunnyvale). Only 32% of the employees at Kawasaki were in indirect labor positions. Of those in indirect labor positions, most were employed in Technology Development.

Profile of American Connector Company

American Connector operated four plants in the U.S. and two in Europe. Each of these plants produced the four basic types of connectors and each serviced a particular customer segment. On the whole, the company's competitive strategy was characterized by its emphasis on both quality and customization. In the marketplace, the company had established a reputation as a high quality supplier; its products were recognized for superior design and performance, in particular. Customers' engineers often commented that American Connector (ACC) provided excellent technical solutions. Within the company itself, quality was a point of pride; management and workers alike believed that quality was American Connector's key to competitive success.

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ACC considered its customization strategy to be an extension of its emphasis on quality. Beyond just meeting manufacturing specifications, quality at ACC meant conforming to the customers' needs. Custom orders made up 15% of the company's total production volume.[7] Employees worked closely with large customers to develop more unique solutions to specific connector problems. For example, engineers at ACC developed a socket for memory chips that made it much easier for end users to later upgrade to higher memory. For arrangements such as this, ACC's designers would collaborate with the customer's engineers early in the development cycle. Many of ACC's custom products had become industry standards.

Historically, ACC had been very profitable, sustaining margins as high as 52%, but management realized that they needed to put forth greater effort to compete globally, increase growth, and maintain profitability in the future. Between 1983 and 1988, the company invested several hundred million dollars worldwide in new plants and equipment to support the strategy. Unfortunately, increased competition in the industry, coupled with slacking demand for connectors made it difficult for the company to reach its growth and profit goals in 1991. While sales had grown from $252 million in 1984 to $800 million in 1991, gross margins had eroded from 52% to 43% during the same period.

The Sunnyvale Plant

American Connector opened the Sunnyvale plant in 1961 in order to serve the emerging electronics-based industries in and around Silicon Valley. Initially, the plant was housed in a small leased facility and had an annual capacity of only one million connectors per year. In 1963, as demand for connectors in the electronics industry began to accelerate, operations were moved to a newly built facility on the site of the present factory. Since that time, the plant was expanded several times in order to accommodate rapidly growing demand. To maintain flexibility, the company tried to expand capacity at the Sunnyvale site ahead of expected growth in demand. Capacity was usually added when long term forecasts indicated that utilization would exceed 85% for sustained periods.[8]

The last major expansion occurred in 1986, bringing capacity up to 600 million units per year. At the time, it was thought that capacity would again have to expanded in 1990 and tentative plans were laid for a completely new factory. However, 1986 proved to be a peak year in the connector market's long period of extraordinary growth. Beginning in 1987, slowed growth in demand resulted in excess capacity industry-wide, so plans for a new factory were immediately halted. Utilization at Sunnyvale plant sunk as low as 50% in 1988, but rebounded to 70% by 1991. Using current demand forecasts, the plant was expected to reach 85% utilization by 1996.

Because of the depressed market conditions, the Sunnyvale plant had made no major investments in capacity or new technology since 1986. This was becoming a primary concern to the plant's production engineering staff. Bob Williams, the Director of Production Engineering explained:

> This plant has always been a technology leader. We've never hesitated to buy the latest production equipment if we thought it could help us improve quality or productivity. I know things have been tight the last few years, but I'm beginning to worry that some of our equipment is no longer leading edge stuff. In the past two years, some nifty new molding presses have hit the market, but our finance people won't let us buy them. Over the long term, we're really going to hurt ourselves if we don't get more aggressive in procuring new equipment.

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[7]This figure did not include prototype production runs which accounted for less than 1% of production.

[8]Sunnyvale calculated capacity utilization assuming a 3¥shift per day, 5¥day per week operation, for 50 weeks of the year. This was the normal operating schedule of the plant.

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The Sunnyvale plant produced the four major types of connectors used in computers, telecommunications equipment, and scientific instruments. Taking into account all of the combinations of housings (which varied by shape and color), the pin configurations, pin platings, and packaging formats, the plant produced about 4,500 different models.

The plant was divided into 5 production areas: terminal stamping and fabrication; terminal plating; plastic housing molding; assembly and testing; and packaging. The typical production flow went as follows. First, a batch of terminals was cut and stamped out of metal wire or strips in the terminal fabrication area. Terminals were then transported to a holding area, where they were kept until they were ready to be plated. Meanwhile, the molding department would fabricate a batch of plastic housings. Given the high rate at which they could be molded, a batch of housings was almost always ready for assembly before the terminal plating operations were completed. Thus, housings were sent to the work-in-process holding area until plated terminals were completed. When terminal plating was completed, the batches of plated terminals and housings were sent to the assembly area where housings and terminals were mated. For most products, Sunnyvale used an automated assembly process. However, for very low volume products (about 10% of Sunnyvale's total volume) manual assembly was required. A completed batch of connectors was then tested and sent to packaging. Sunnyvale offered a very wide range of packaging formats, ranging from a 10-piece plastic bag to 1,500 piece loaded reel. The processing lead time for a batch of connectors was typically 10 days for standard items and two to three weeks for special order items. Some runs ran as long as one week, but most product lines were run for 1.5 to 2 days. Sunnyvale maintained a finished goods inventory of 38 days.

Each area had its own production supervisor who reported to the plant's Director of Manufacturing. The key responsibility of area supervisors was to ensure that their respective areas met the production schedule. The Production Control Department (PCD) was responsible for scheduling the plant's aggregate production and for coordinating production across the five production areas. Using marketing forecasts and data from the finance department, the PCD set an annual production schedule which specified monthly production targets. The total level of production and the mix of products for any given month was updated three months in advance and each production area was advised. The production schedule for any given day was supposed to be "frozen" thirty days in advance. Thus, for example, on October 15 each area would receive its exact production schedule for November 15. In reality, however, the schedule was routinely changed to accommodate rush orders and requests from important customers. It was not unusual for some schedule changes to occur on a weekly and even daily basis. While such changes in the schedule could be disruptive, most of the supervisors in the plant understood the importance of maintaining flexibility. Brad Wornham, Director of Production Control, noted:

> We've always prided ourself on being responsive to customers' needs and sometimes that means changing your production schedule at the last minute. You have to understand that many of our orders are for connectors which our engineering group designed specifically for a customer's new model. It's almost impossible for our customers to predict demand for their products with any degree of certainty. If they wind up with a real winner, we've got to be able to supply it or they can take the design to someone else. When that happens, our engineering people and our marketing people go crazy.
>
> Sure, I wish we could get better information from our big customers. But, realistically, we can't demand it. If you want to compete in this business, particularly in today's market, you've got to be able to meet customers' delivery requirements. The guy who had this job before me believed that the schedule was sacred and should never be changed. He was always butting heads with marketing and engineering. In the end, marketing and engineering won and he has, as they say, moved on to other opportunities.

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In recent years, Production Control had come under increasing pressure to schedule production runs in a way that would reduce work-in-progress (WIP) inventory. In the past, high WIP was not viewed as a problem because any extra inventory carrying costs were easily covered by growing sales. In the current climate, however, excessive WIP was considered a burden that the plant could ill afford to carry. Brad Wornham explained:

> We've always kept a good amount of WIP in this factory. The set-up times in molding and plating operations don't make it economical to produce every one of our products in the exact quantity required every month. Instead, molded housings and terminals for every one of our products is produced in some minimum efficient batch size. Whatever parts are not needed in assembly that month stay in the WIP stockroom until they are needed.
>
> I know finance is getting worried about the costs of all this WIP, but it buys us some flexibility. We don't like to keep finished goods inventory because of obsolescence. But if we keep housings and terminals in WIP, we can respond quickly to unexpected orders without keeping a lot of finished goods. Finance has suggested we try shorter production runs, but they don't like what that does to our utilization figures. Our average run in molding is already down to 1.5 days. If we go any shorter, it's really going to affect costs.

In recent years, production scheduling had become an increasingly stressful task as the number of individual products manufactured at Sunnyvale expanded. In 1984, the plant produced about 3,000 different connectors; this figure had risen to 3,500 in 1986 and to 4,500 in 1991. To keep on top of the increasingly complex production schedule, the Production Control staff had expanded steadily over the past 6 years from 42 to 65. Some of the pressure on the staff was also relieved in 1988 when the plant invested $500,000 in a new computer system and software for production scheduling.

Quality had been identified as an area offering major opportunities for improvement. As a company, American Connector had an outstanding reputation for quality. As Bob Williams put it: "Our customers' engineers love us because of our superior design and their production people love us because we deliver nearly perfect connectors and never miss a scheduled delivery." However, some within ACC were becoming concerned about the way Sunnyvale was pursuing quality. Within the plant, defect rates were relatively high (about 26,000 per million units of production in 1990) and final inspection was responsible for making sure these didn't reach the customer. As one test technician described: "When it comes to quality, we do it the old fashioned way—we inspect it." In recent years, the new manager of the Quality Control department had made some attempts to implement Statistical Process Control and other more defect preventive measures, but these efforts had made only minimal progress. Some of the production supervisors felt that many of the defects were a result of new product designs which used increasingly complex pin configurations and required extremely high tolerances for molding. Yields on newly designed products entering production for the first time were sometimes as low as 55%. However, yields typically improved to about 98% once a product was in production for at least one year.

The Options

As she reviewed the data, Denise Larsen contemplated different scenarios and responses. She asked both Jack Mitchell and Andrew Li for their opinions. Jack Mitchell commented:

> From what I can see, DJC will kill us if they can operate a plant like Kawasaki in the U.S. We need to completely change our approach to manufacturing if we intend to stay competitive. We can't afford to wait and see if DJC will build a plant here or

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> how it will work—we need to strike first and model our own operation into one like Kawasaki. There is no reason we can't do it here before they do.

Andrew Li looked skeptical. He responded:

> I don't see any need to panic. It's not going to be easy for DJC to implement their operating strategy in the U.S. I'm willing to bet that their new plant here will be completely different from the Kawasaki plant. In any case, we already have a team working on a plan to cut costs. They will have a formal proposal ready in a few days. Before we do anything radical, let's look at the proposal and consider whether it will work.

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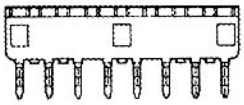
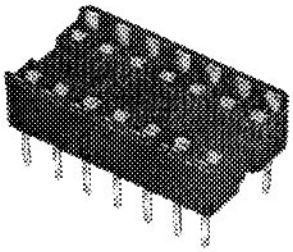
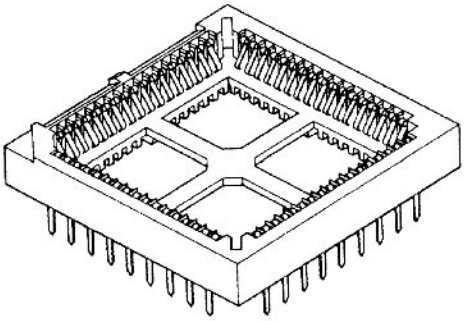
Exhibit 1 A Sample of Connector Products by Type

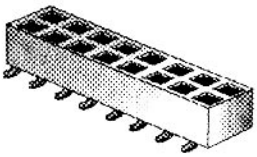
Figures 1-3: Component/Chip-to-Board Connectors

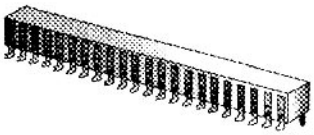


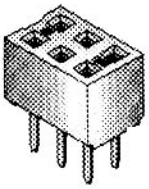




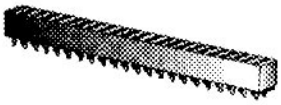
Figures 4-6: Board-to-Board Connectors







Figures 7-8: Wire-to-Board Connectors



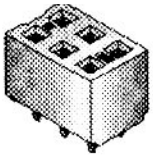


Figure 9: Wire-to-Wire Connectors

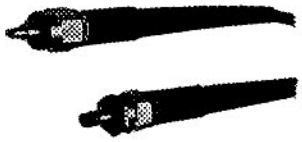


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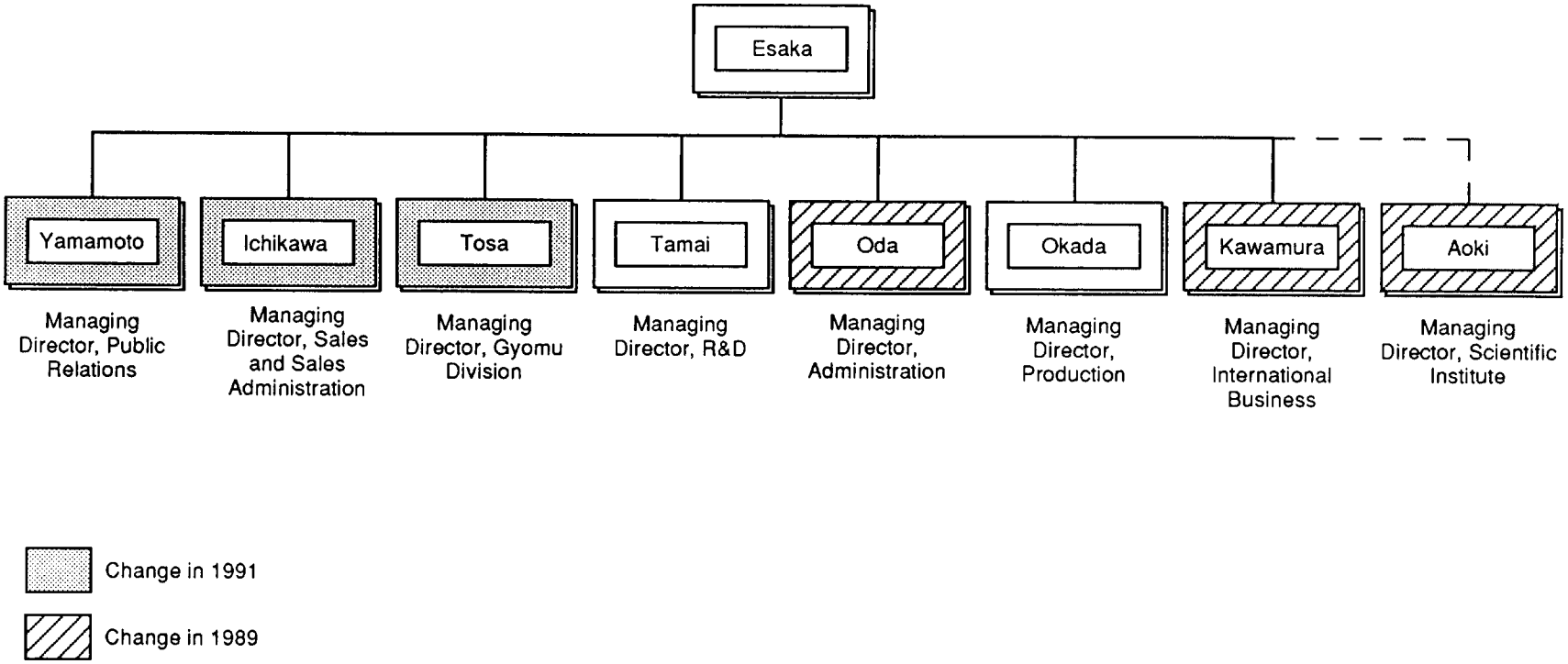
Exhibit 2 DJC Corporation: Organizational Chart (1991)



Source: Company Financial Reports

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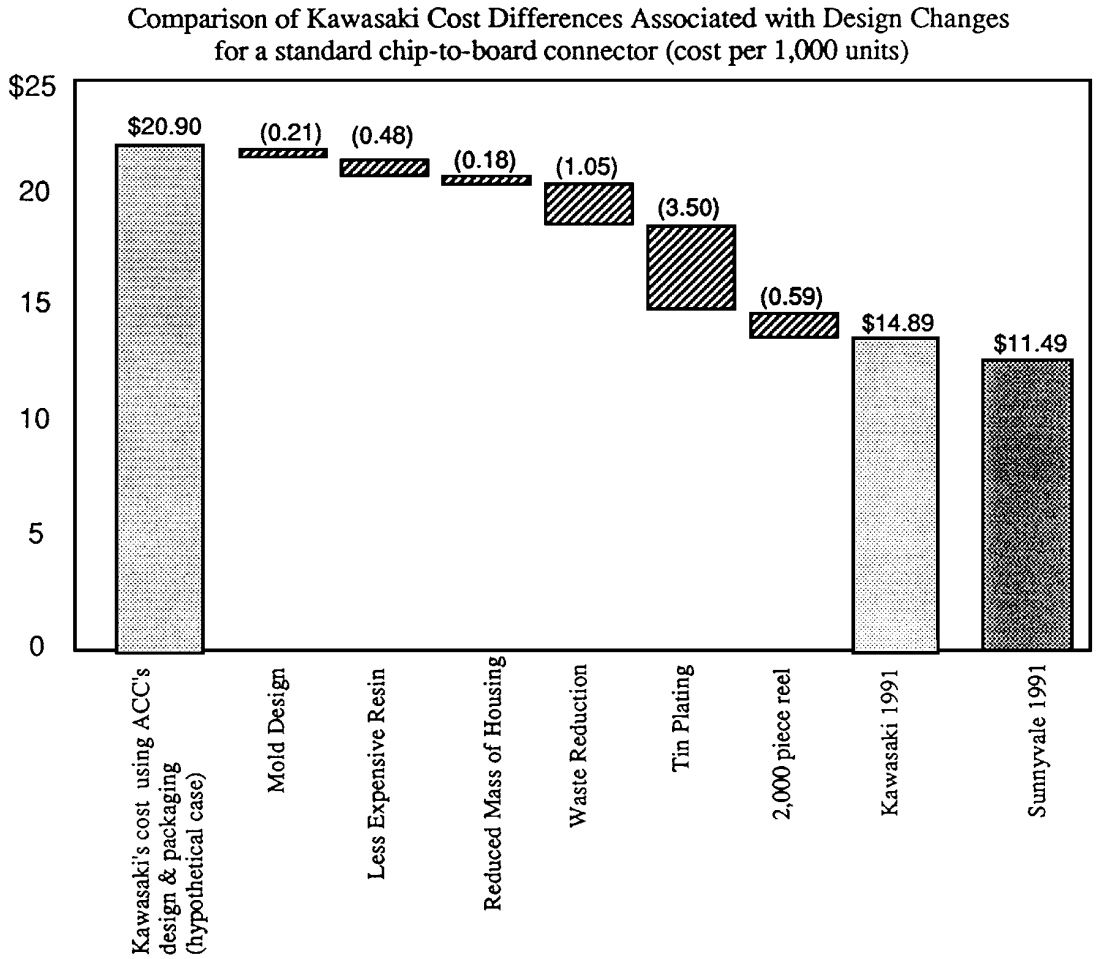
Exhibit 3 Analysis of Kawasaki's Material Cost Savings, Given American Connector's Design



Explanation of Design Changes which resulted in Material Cost Savings

• Mold Design: By using shorter and narrower "runners" (i.e. the channels which carried plastic resin into the mold chamber), DJC was able to reduce the consumption of raw material in molding.

• Less Expensive Resin: DJC's engineering department found that certain expensive resins used in plastic housings could be substituted with much lower cost materials, without any significant degradation in product performance, reliability, or durability.

• Reduced Mass of Housing: DJC altered connector designs (e.g. thinner walls) so that any unnecessary mass in the plastic housing could be eliminated.

• Waste Reduction: Includes material costs saved by reducing defects in both plastic housing molding and terminal stamping, as well as damage incurred during material handling.

• Tin Plating: Whenever possible and whenever it would not reduce product performance, DJC plated terminals with tin rather than with gold or other higher cost metals.

• 2,000 Piece Reel: DJC packed 2,000 pieces onto a packaging reel (instead of 1,500). This resulted in lower consumption of reels.

Note: Cost changes represent difference in material costs only, does not include manufacturing efficiencies.
Source: Company Documents

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Exhibit 4 Organizational Chart of Kawasaki Plant

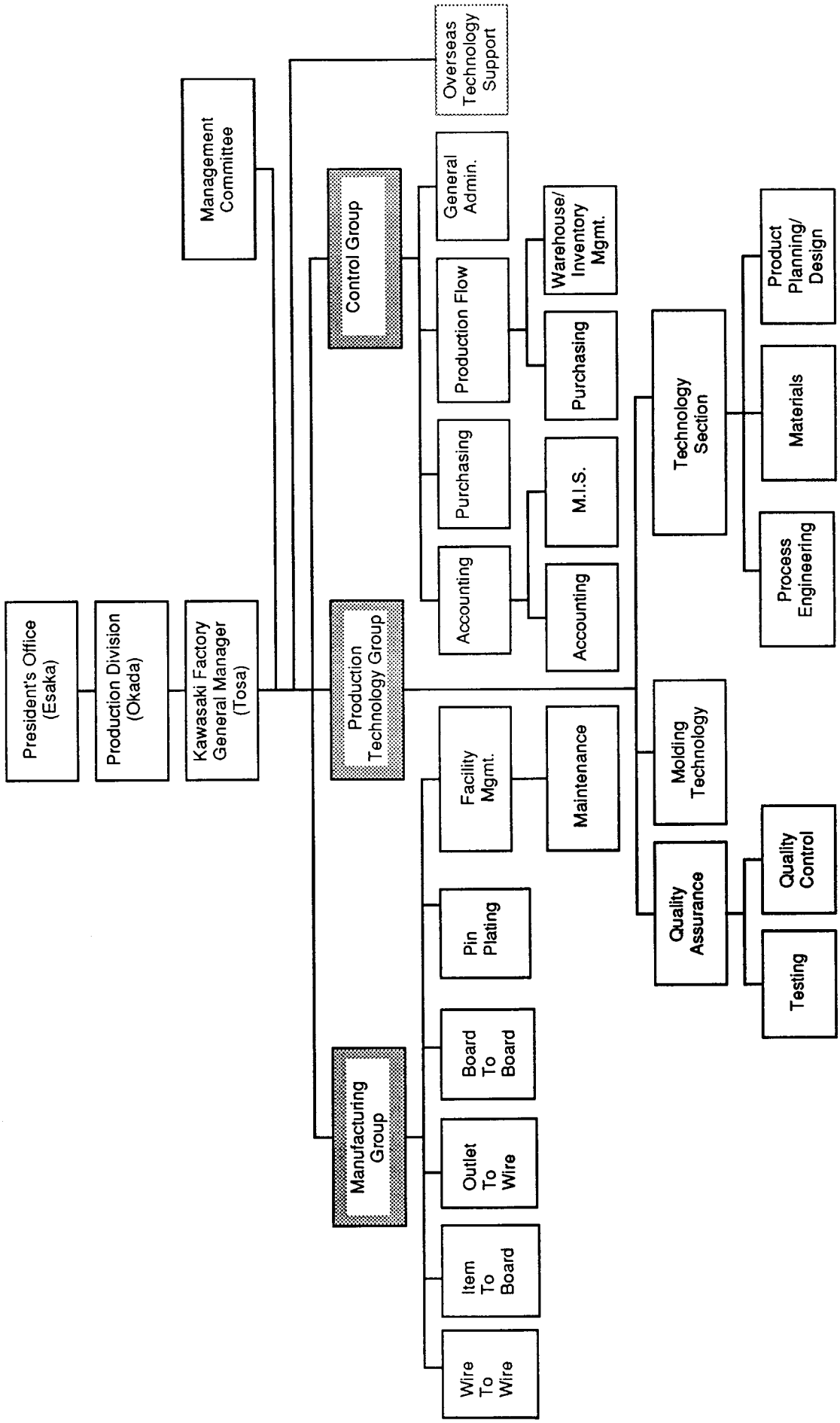

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Exhibit 5 Comparison of Labor Use (1991)

	Kawasaki 94 Employees	Sunnyvale 396 Employees
Indirect Labor		
Control	11.7%	16.7%
Technology Development	12.8%	6.8%
Materials Handling	3.2%	10.4%
Mechanics	4.3%	11.9%
Direct Labor (Production)	68.0%	54.0%
Total	100.0%	100.0%

Note: Kawasaki production estimated to be 700 million units.
Sunnyvale production estimated to be 420 million units.

Exhibit 6 Productivity Comparisons (1991)

employees	Kawasaki 94 employees	Sunnyvale 396 employees
(a) Connector Output per Square Foot (in thousands of units) Per Square Foot of Map Space	15.1	10.9
(b) Connector Output per Employee (in millions of units)	7.45	1.06
(c) Fixed Asset Utilization (%)		
Plant Not Operating*	5.7%	28.6%
Non-scheduled	13.2%	23.5%
Process Failure	1.0%	8.9%
Preventive Maintenance	2.0%	2.4%
Process Changeover	2.0%	4.8%
Quality Losses	0.7%	1.6%
Effective Utilization	75.4%	30.2%

* Assumes maximum available time of 24 hours/day, 350 days/year (8,400 hours). Equal to (350 - days plant is operating)/350.

Source: Company documents

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Exhibit 7 Comparison of Manufacturing Costs (Kawasaki vs. Sunnyvale)

DJC vs. American Connector
Cost of Goods Sold
For a Standard Chip-to-Board Connector
(dollars per 1,000 units)

	DJC/Kawasaki 1986	ACC/Sunnyvale 1986	DJC/Kawasaki 1991	ACC/Sunnyvale 1991
Raw Material, Product	14.32	10.40	12.13	9.39
Raw Material, Packaging	3.27	2.25	2.76	2.10
Labor, Direct	7.63	...	3.02	...
Labor, Indirect	2.30	...	.75	...
Total Labor	...	8.53	...	10.30
Electricity	2.47	1.80	1.40	.80
Depreciation	7.63	5.52	1.80	5.10
Other	4.12	4.41	4.24	6.10
TOTAL	41.74	32.91	26.10	33.79

Note: -135 = $1, 1989 exchange rate used to convert from yen to dollars.

Exhibit 8 Cost Indices, United States/Japan (1991)

Expense Item	Index
Raw Material, Product	.60
Raw Material, Packaging	.60
Labor, Direct	1.10
Labor, Indirect	1.10
Electricity	.80
Depreciation	1.00
Other	1.00

Source: Company documents

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